FINITE PRESENTATION SPACES OVER A LIMIT FORM THE COLIMIT CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2161-2227, following the statement at 2147-2159.

Editorial note. The single '(details omitted)' covers finite presentation of W over U and of R' over R: etale maps are locally of finite presentation and fp is preserved under base change; flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 2179 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Affine etale cover of a quasi-compact algebraic space. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-affine-cover (cross-chapter, cited at ~2163-2164). Produces the affine etale U_0 -> X_0.
- Presentation of the limit space. lemma-directed-inverse-system-has-limit (same chapter, cited at ~2168-2169). Gives X = U/R with U = lim U_i and R = lim R_i.
- Descent of finite presentation along affine limits of schemes. Limits, Lemma limits-lemma-descend-finite-presentation (cross-chapter, cited at ~2180 and ~2190). Descends W -> U and R' -> R to some finite level; applied twice.
- Descent of etale maps and monomorphisms. Limits, Lemmas limits-lemma-descend-etale and limits-lemma-descend-monomorphism (cross-chapter, cited at ~2193-2194). Makes s'_i, t'_i etale and j'_i a monomorphism after increasing i.
- Presentation theorem for algebraic spaces. Spaces, Section spaces-section-presentations and Theorem spaces-theorem-presentation (cross-chapter, cited at ~2177 and ~2199). Y = W/R' and Y_i = W_i/R'_i are algebraic spaces.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SL: focused-seeds/geometry/S9-spaces-limits-descend-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction

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Source reference. Original reference 07SK: focused-seeds/geometry/S9-spaces-limits-descend-finite-presentation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SM: focused-seeds/geometry/S9-spaces-limits-descend-finite-presentation/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 04AK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 694–694. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H6: focused-seeds/geometry/S9-spaces-limits-descend-finite-presentation/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0261: focused-seeds/geometry/S9-spaces-limits-descend-finite-presentation/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02WW: focused-seeds/geometry/S9-spaces-limits-descend-finite-presentation/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let I be a directed set. Let $(X_i, f_{ii'})$ be an inverse system over I of algebraic spaces over S. Assume*

1. *the morphisms $f_{ii'} : X_i \to X_{i'}$ are affine,*
2. *the spaces X_i are quasi-compact and quasi-separated.*

Let $X = \lim_i X_i$. Then the category of algebraic spaces of finite presentation over X is the colimit over I of the categories of algebraic spaces of finite presentation over X_i.

Proof. Pick $0 \in I$. Choose a surjective étale morphism $U_0 \to X_0$ where U_0 is an affine scheme (Properties of Spaces, Lemma 03H6). Set $U_i = X_i \times_{X_0} U_0$. Set $R_0 = U_0 \times_{X_0} U_0$ and $R_i = R_0 \times_{X_0} X_i$. Denote $s_i, t_i : R_i \to U_i$ and $s, t : R \to U$ the two projections. In the proof of Lemma 07SF we have seen that there exists a presentation $X = U/R$ with $U = \lim U_i$ and $R = \lim R_i$. Note that U_i and

U are affine and that R_i and R are quasi-compact and separated (as X_i is quasi-separated). Let Y be an algebraic space over S and let $Y \to X$ be a morphism of finite presentation. Set $V = U \times_X Y$. This is an algebraic space of finite presentation over U. Choose an affine scheme W and a surjective étale morphism $W \to V$. Then $W \to Y$ is surjective étale as well. Set $R' = W \times_Y W$ so that $Y = W/R'$ (see Spaces, Section 0261). Note that W is a scheme of finite presentation over U and that R' is a scheme of finite presentation over R (details omitted). By Limits, Lemma 07SK we can find an index i and a morphism of schemes $W_i \to U_i$ of finite presentation whose base change to U gives $W \to U$. Similarly we can find, after possibly increasing i, a scheme R'_i of finite presentation over R_i whose base change to R is R'. The projection morphisms $s', t' : R' \to W$ are morphisms over the projection morphisms $s, t : R \to U$. Hence we can view s', resp. t' as a morphism between schemes of finite presentation over U (with structure morphism $R' \to U$ given by $R' \to R$ followed by s, resp. t). Hence we can apply Limits, Lemma 07SK again to see that, after possibly increasing i, there exist morphisms $s'_i, t'_i : R'_i \to W_i$, whose base change to U is S', t'. By Limits, Lemmas 07SL and 07SM we may assume that s'_i, t'_i are étale and that $j'_i : R'_i \to W_i \times_{X_i} W_i$ is a monomorphism (here we view j'_i as a morphism of schemes of finite presentation over U_i via one of the projections – it doesn't matter which one). Setting $Y_i = W_i/R'_i$ (see Spaces, Theorem 02WW) we obtain an algebraic space of finite presentation over X_i whose base change to X is isomorphic to Y.

This shows that every algebraic space of finite presentation over X comes from an algebraic space of finite presentation over some X_i, i.e., it shows that the functor of the lemma is essentially surjective. To show that it is fully faithful, consider an index $0 \in I$ and two algebraic spaces Y_0, Z_0 of finite presentation over X_0. Set $Y_i = X_i \times_{X_0} Y_0$, $Y = X \times_{X_0} Y_0$, $Z_i = X_i \times_{X_0} Z_0$, and $Z = X \times_{X_0} Z_0$. Let $\alpha : Y \to Z$ be a morphism of algebraic spaces over X. Choose a surjective étale morphism $V_0 \to Y_0$ where V_0 is an affine scheme. Set $V_i = V_0 \times_{Y_0} Y_i$ and $V = V_0 \times_{Y_0} Y$ which are affine schemes endowed with surjective étale morphisms to Y_i and Y. The composition $V \to Y \to Z \to Z_0$ comes from a (essentially unique) morphism $V_i \to Z_0$ for some $i \geq 0$ by Proposition 04AK (applied to $Z_0 \to X_0$ which is of finite presentation by assumption). After increasing i the two compositions

$$V_i \times_{Y_i} V_i \to V_i \to Z_0$$

are equal as this is true in the limit. Hence we obtain a (essentially unique) morphism $Y_i \to Z_0$. Since this is a morphism over X_0 it induces a morphism into $Z_i = Z_0 \times_{X_0} X_i$ as desired. □